



Problem 2

- i) No, the given ANN is not suitable for multiclass classification. The ANN contains only 1 output neuron with sigmoid activation function.

A single output neuron can produce only one probability value between 0 and 1, which is suitable for binary classification problems such as Yes/No or True/False predictions. In multiclass classification, the model must classify the input into one among multiple classes, which cannot be efficiently represented using only one output neuron.

For multiclass classification, the network should contain multiple output neurons, where each neuron represents one class.

Therefore, to make this ANN suitable for multiclass classification:

- The number of output neurons should be equal to the number of classes.
- Sigmoid activation should be replaced with softmax activation.
- Categorical cross-entropy loss should be used instead of binary prediction methods.



If the network has 2 output nodes instead of 1, the output layer will produce two predicted outputs:

y_1 and y_2

Each input output neuron will have its own weighted sum and activation function.

for output node 1:

$$Z_{op1} = w_{1op1} H_1 + w_{2op1} H_2 + w_{3op1} H_3 + w_{op1}$$

$$y_1 = \sigma(Z_{op1})$$

for output node 2:

$$Z_{op2} = w_{1op2} H_1 + w_{2op2} H_2 + w_{3op2} H_3 + w_{op2}$$

$$y_2 = \sigma(Z_{op2})$$

The total loss function becomes the sum of losses from both output neurons.

$$L = \frac{1}{2} (y_1 - \hat{y}_1)^2 + \frac{1}{2} (y_2 - \hat{y}_2)^2$$

Thus, adding 2 ~~output~~ output nodes means:

- Separate weights and biases for each O/P neuron.
- Separate predicted o/p's
- Combined loss from both outputs during training.



Q3) If binary cross-entropy (BCE) loss is used instead of mean squared error (MSE), then only the loss function and its derivative will change. The forward propagation equations remain the same.

The Binary cross-entropy loss function is given by:

$$L = - [y \log(y) + (1-y) \log(1-y)]$$

where

y = actual output

$\hat{y}$ = predicted output

for sigmoid activation with BCE loss, the gradient at the output layer simplifies to:

$$\frac{\partial L}{\partial z_{op}} = (y - \hat{y})$$

Therefore, the gradient for output layer weights becomes:

$$\frac{\partial L}{\partial w_{io}} = (y - \hat{y}) h_i$$

Similarly, the gradient for hidden layer weights connected to h_3 becomes

$$\frac{\partial L}{\partial w_{i3}} = (y - \hat{y}) w_{3op} h_3 (1 - h_3) x_i$$



Compared to MSE loss, BCE loss removes the extra sigmoid derivative term at the output layer, making training faster and more efficient for binary classification problems.